

AI-Agent Project Rules

A working agreement for building a project step-by-step with an AI coding agent

1. Work in Small, Sequential Steps

Never let the agent build the whole project in one shot. Break the project into a numbered list of steps before writing anything, then execute one step at a time.

- Agree on the full step list first (a roadmap), then start Step 1 only.
- One step = one small, testable unit of work (a feature, a fix, a file).
- Do not move to the next step until the current one is confirmed working.
- If a step turns out to be too big, split it into sub-steps instead of pushing through.

2. No Unnecessary Code

Code should only appear when it is actually being written to a file for the current step. Avoid code dumps, alternate versions, or "just in case" snippets.

- No full-file reprints when only a few lines changed — show/apply only the diff.
- No speculative code for features not yet reached in the roadmap.
- No duplicate implementations ("here's another way to do it") unless asked for.
- Explanations should be short; let the working code and its output speak for themselves.

3. Explain Before Executing

Before each step is built, the agent should state in one or two lines what it is about to do and why — not a code wall, just the plan for that step.

- State the goal of the step, the files it will touch, and any assumption being made.
- Ask before doing anything destructive: deleting files, overwriting configs, changing schemas.
- If the request is ambiguous, the agent should pick the most sensible default and say so, rather than stall.

4. Maintain a Living Project Document

Keep one running document (e.g. **PROJECT.md** or **PROGRESS.md**) that is updated after every step. This is the single source of truth for where the project stands.

The document should track:

- **Project overview** — goal, tech stack, key decisions and why they were made.
- **Step log** — each completed step, what changed, and the date/version.
- **Current status** — what's done, what's in progress, what's next.
- **Open issues / known bugs** — anything unresolved, so nothing is forgotten.

- **File map** — short description of what each major file/folder does.
- **How to run it** — setup and run instructions, kept current.

Rule of thumb: if a step changes the project in any way, the document is updated in the same turn — not batched up for later.

5. Confirm Before Moving Forward

- After each step, briefly state what was done and how to verify it (run/test instructions).
- Wait for a go-ahead (or visible confirmation the step works) before starting the next step.
- If something breaks, fix it in the current step before advancing — don't stack new work on a broken base.

6. Version Control Discipline

- One logical step = one commit, with a clear, specific commit message.
- Never mix unrelated changes (a feature + a random refactor) in the same step/commit.
- Keep a **CHANGELOG** section (can live inside the project document) for anything shipped.

7. Keep Scope Under Control

- Stick to what was agreed in the roadmap; flag scope creep instead of silently absorbing it.
- New ideas go into a "Backlog / Future Ideas" section of the project document, not into the current step.
- Prefer the simplest solution that works over a clever or over-engineered one.

8. Testing and Verification

- Each step should include a way to verify it: a test, a manual check, or sample input/output.
- Don't mark a step "done" without it actually being run/verified at least once.
- Edge cases and error handling are called out explicitly, not assumed.

9. Context and Memory Hygiene

- Reference the project document instead of relying on memory of earlier chat turns.
- If context is getting long or is at risk of being lost (new session, model switch), re-read the project document first before continuing.
- Keep secrets/API keys out of the document and out of chat — use placeholders and environment variables.

10. Communication Style

- Plain, direct updates: what changed, why, what's next — no filler.

- Flag risks, trade-offs, and assumptions instead of silently picking one and moving on.
 - If a request conflicts with an earlier decision in the project document, say so before proceeding.
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Suggested PROJECT.md Template

Project Name

Goal: ...

Tech stack: ...

Status

Current step: ... | Last updated: ...

Step Log

- Step 1 — [date] — what was built, key decisions
- Step 2 — [date] — ...

File Map

- /src/... — description

Open Issues

- ...

Backlog / Future Ideas

- ...

How to Run

- ...

Golden rule: one small step, one visible result, one updated document — every time, before moving on.